STA 2002, Fall 2024
Probability & Statistics II

Introduction

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Tutorials & Homework

- **Tutorials start on the second week, the week of Sep 9.**
- **The class has weakly homework assignment. Each homework set will have about 2-3 questions, and 1 programming assignment.**
- **Electronic submission on blackboard (https://bb.cuhk.edu.cn)**
- **Homework will be given on each Sunday, and deadline will be 11:59p.m., next Sunday.**
- **No late homework will be accepted**

Learning Assessment Strategy

5. Assessment Scheme

Component/ method	% weight
Assignments	25
Mid-Term Exam	35
Final Exam	40

- **Midterm (tentative): TBD, will be on 8th week, the week of Oct 21.**
- **There will be no make-up exams**
- **Zero tolerance policy when it comes to cheating so, DON'T CHEAT**
- **Missing the midterm or final exam without prior notification to and approval of the instructors will automatically result in the "0" grade for the exam.**

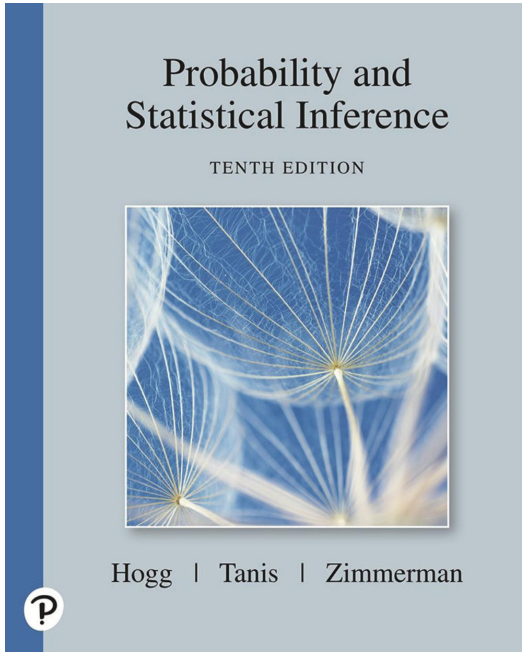
Resources

- **Textbook: Probability and Statistical Inference**

  by R.V. Hogg, E.A. Tanis, and D.L. Zimmerman

  Lecture Notes

- **Blackboard site: https://bb.cuhk.edu.cn**



Tentative Calendar

10. Indicative teaching plan

Week	Content/ topic/ activity
1 - 2	General Introduction. Review of probability basics and families of distributions. Descriptive statistics.
11-12	Simple Linear Regression. More tests concerning regression and correlation.
13-14	Likelihood ratio tests. Bayesian estimation. Final Review.
3 - 4	Point estimation (Maximum Likelihood Estimation and Method of moments).

	Confidence interval (CI) estimation for means; CI for the difference of two means; CI for proportions. Sample size.
5 - 6	Tests of statistical hypotheses, Critical region, Significance level, p-value, Power, t-test, F-test. Tests about proportions.
7	More tests (Sample size; Power of a statistical test).
8	Review Session and Midterm Exam
9 - 10	Distribution-free CIs and tests. Chi-square goodness of fit. Contingency tables. ANOVA.

1 Introduction

1.1 Statistical Inference

Reading(s) 1.1. Wasserman (2004) Section 6.1 and 6.2.

Statistical *inference*, informally speaking, is to make sense of data with certain *model*. This includes, but not limited to, the following aspects:

- identify *interpretable* patterns of the data
- help scientific endeavors by comparing scientific theory with empirical evidence, and planning experimental observations.

Statistical philosophy is that some *unknown* probability measures and/or processes generate the data, and statistical inference is to describe and understand the underlying data generating process. To this end, we define a statistical model as a set of distributions.

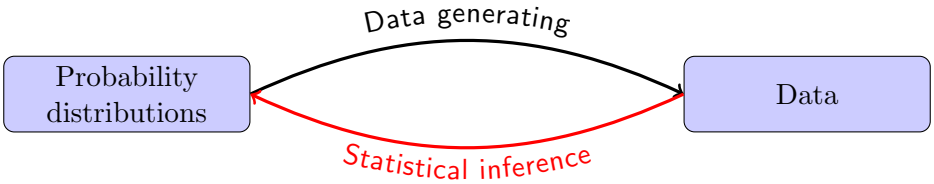


Figure 1: Statistical model and data

Parametric and nonparametric models Statistical models are generally classified into parametric models and non-parametric models. There are also semi-parametric models that have both parametric and non-parametric specification in the model.

A *parametric* model is a set of distributions that can be parameterized by a finite number of parameters. For instance, we could obtain a random sample from some univariate Gaussian distribution with unknown mean and variance. Here, the parametric model is univariate Gaussian, and the parameters are the mean and the variance. If you are *only* intersted in the mean, then the mean parameter is called *the parameter of interest* and the variance parameter is called a *nuisance* parameter.

A *non-parametric* model is a set of distributions that cannot be parameterized by a *fixed* number of parameters. For instance, in density estimation, the class of all pdf's cannot be parameterized by a fixed number of parameters, hence it is non-parametric. Similarly, in estimating CDF, the class of all CDF's is also non-parametric. In the regression setting, flexible regression functions are also considered as nonparametric, including splines, local polynomials, deep neural networks with ReLU activation function, and so on. Models that have a growing number of parameters as sample size grows are also called non-parametric, e.g., Bayesian nonparametric methods with Dirichlet process (mixture) priors.

1.2 The Box's Loop

Reading(s) 1.2. Blei (2014) and Box (1976).

Why *should* we study statistical inference? Statistical inference is the core and fundamental building block of modern data science.

Figure 2 presents a big picture of modern data analysis procedure/pipeline. The flow chart is essentially the Box's loop (Box, 1976; Blei, 2014). Box's loop illustrates the iterative nature of statistical analysis. Typically we cannot finish data analysis by fitting a single model. Instead, we strive to provide statistical analysis as accurate and convincing as possible by exploring various modeling options and alternatives.

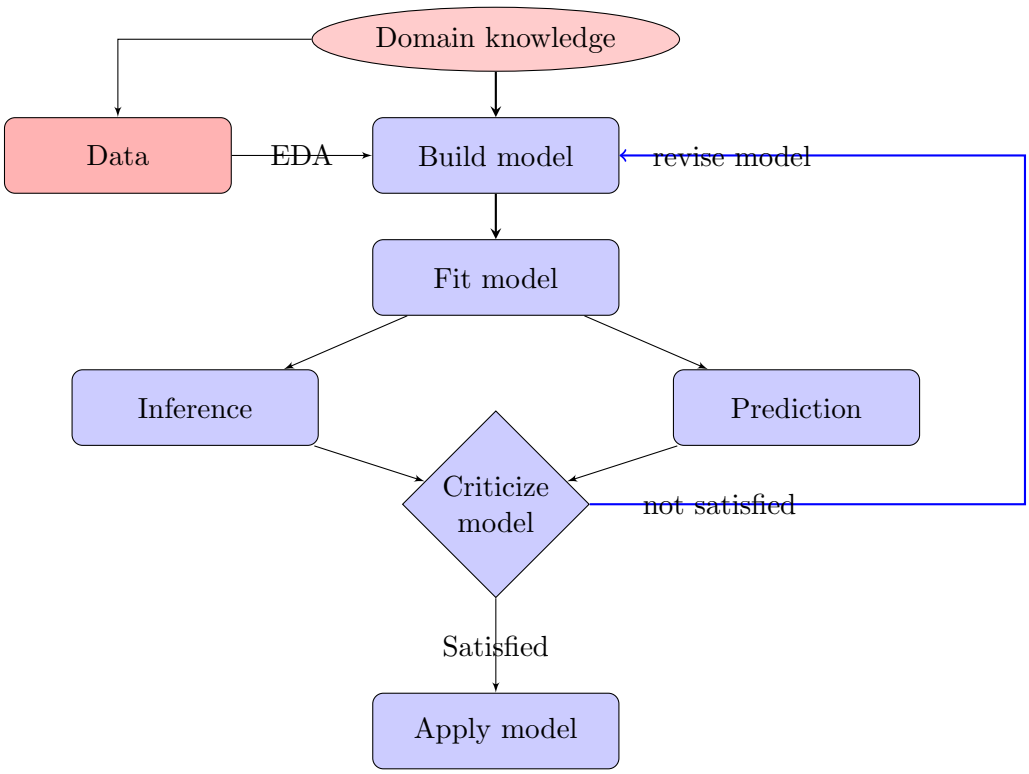


Figure 2: Box's Loop

Often times, statisticians/data scientists work with some domain experts. Data is often collected and cleaned under the guidance of certain domain knowledge.

We first summarize, visualize, and hypothesize about the data, using scatter plots, histograms, and other tools. At this stage, we also check the missingness, censorship and truncation of the data. This step is called exploratory data analysis (EDA).

After EDA, we build a probabilistic model with both domain knowledge and EDA results.

After model fitting, we make inference about the hidden quantities of the model and/or make predictions with the current model.

With the inference and/or prediction, we criticize the model by examining the model fit, checking the validity of the inference, and others. In assessing the model fit, we can ask the following questions,

- are we missing important features of the data?
- does our model work reasonably well on hold-out sample?
- is there a better way to model the data?

In checking the validity of the statistical analysis, we can ask the following questions:

- are the conclusions sensible to the domain experts?
- what are the important ways that the current model succeeds and fails?

If we are not satisfied with the current model, we revise it and repeat the steps of model fitting and criticism.

If we are satisfied with the current model, we can use this model to answer questions and make predictions.

Another important feature of modern data analysis is its collaborative nature. As more and more statistical tools are developed, domain experts may encounter data that are generated from a complicated process, but it might not be so efficient learning all the statistical tools themselves. A natural and more efficient alternative is that the domain experts come to statisticians with certain dataset and questions in mind. So the statisticians need to communicate with the domain experts on their goal(s) of inference and the data generating process.

1.3 Fundamental Tasks of Statistical Inference

Reading(s) 1.3. Wasserman (2004) Section 6.3.

We use parametric models to illustrate the fundamental concepts of statistical inference. Suppose the model is

$$\{\mathbb{P}_\theta : \theta \in \Theta\},$$

where Θ is the *parameter space*, and the observed data is independently generated from $\mathbb{P}_{\theta_0}$ where θ_0 is some "true" value. Our major interest is to learn θ_0.

1.3.1 Point Estimation

Point estimation of θ_0 is to provide an algorithm, say $t(\mathbf{X})$, such that the output is an estimate of θ_0. The output of the algorithm is data dependent, and the output is a number or vector that matches the dimension of θ_0. Since the output is essentially a point in Euclidean space, we call it point estimate. The conventional notation for point estimates is the *hat* notation: $\hat{\theta} = t(\mathbf{X})$.

Example 1.1 (Normal mean)**.** The model is $X_1, ..., X_n \overset{iid}{\sim} N(\theta, 1)$, and the goal is to provide a point estimator for the mean parameter θ.

Firstly, $\hat{\theta}$ could be $\frac{1}{n}\sum_{i=1}^n X_i$, the sample average. Secondly, $\hat{\theta}$ could be sample median. Thirdly, $\hat{\theta}$ could be X_1, the very first observation. There could be countless point estimators, some of which are more reasonable and "better" than others. ♢

Example 1.2 (Prediction)**.** Model-based prediction tries to say something beyond what we have observed with the statistical model. Continuing the Normal mean example, we can predict the next value to be observed with (some function of) the point estimate of θ. Its optimality requires decision-theoretic argument, which is out of the scope of this course. Essentially, we treat the next observation as an unknown parameter which can be a function of the model parameters. ♢

1.3.2 Confidence Sets

Point estimates are expected to be off the target in most cases, that is, the probability that the point estimate happens to be exactly the true value is 0:

$$\mathbb{P}_{\theta_0}(t(\mathbf{X}) = \theta_0) = 0.$$

A reasonable alternative is to replace the point estimate with a interval/set: $C(\mathbf{X})$. Like point estimation, interval/set estimation is to provide an algorithm, say $C(\mathbf{X})$, such that the output is a interval/set, with the hope that the interval/set contains the true value.

The inferential statement of the confidence interval/set is that the true value is in the interval/set $C(\mathbf{X})$ with at least $(1-\alpha)$ confidence:

$$\mathbb{P}_{\theta_0}(\theta_0 \in C(\mathbf{X})) \geq 1 - \alpha,$$

where we call $(1-\alpha)$ as the *coverage* of the confidence interval $C(\mathbf{X})$.

The interpretation of the confidence interval/set is a bit tricky. Firstly, despite the confidence interval/set is data dependent, for a particular dataset, the confidence interval/set is fixed. That is,

for a particular dataset, the confidence interval/set either contains the true value or does not contain it. Unfortunately, it is often impossible to know which scenario we are in.

Secondly, the probability $\mathbb{P}_{\theta_0}(\theta_0 \in C(\mathbf{X}))$ is approximately the relative frequency of the confidence interval containing the true value, if we *repeat* the data generating process/experiment sufficiently many times. That is, despite we cannot say much about the parameter θ given a particular dataset, the procedure of interval/set estimation is guaranteed to cover the true value with certain probability, even before any data is generated and observed. The "confidence" of the confidence interval/set is procedure-wise.

Example 1.3 (Bernoulli)**.** Suppose $X_1, ..., X_n \overset{iid}{\sim} Ber(\theta)$. For certain values of θ_0, it is easy to show $\mathbb{P}_0(\hat{\theta} = \theta_0) = 0$. We can use the confidence interval $C(\boldsymbol{X}) = (\hat{\theta} - 2\sqrt{\frac{1}{n}\hat{\theta}(1-\hat{\theta})}, \hat{\theta} + 2\sqrt{\frac{1}{n}\hat{\theta}(1-\hat{\theta})})$, and it is easy to see that $\mathbb{P}(\theta_0 \in C(\boldsymbol{X})) > 0$. The exact coverage rate of $C(\boldsymbol{X})$ is often very challenging. $\diamondsuit$

1.3.3 Hypothesis Testing

Hypothesis testing addresses a slightly different problem than estimation. The focus is to evaluate the validity of certain statement which is called hypothesis. The hypothesis is typically formulated in terms of the parameter. The vanilla case of hypothesis testing has two hypotheses: the null hypothesis denoted as H_0, and the alternative hypothesis denoted as H_1.

For instance, we can have the following hypothesis testing problem:

$$H_0 : \theta_0 \geq 0, v.s.\ H_1 : \theta_0 < 0.$$

With certain reparameterization, such hypothesis testing problem is called A/B test. More generally, we can test the hypotheses

$$H_0 : \theta_0 \in \Theta_0, v.s.\ H_1 : \theta_0 \in \Theta_1.$$

In many genetics studies, we want to test a large collection of hypotheses all at once:

$$H_{0j} : \theta_{0j} \in \Theta_{0j}, v.s.\ H_{1j} : \theta_{0j} \in \Theta_{1j} \text{ for } j = 1, ..., N,$$

where j is the index of certain gene, and N can be tens of thousands.

1.4 The Paradigms of Statistical Inference

Reading(s) 1.4. Efron and Hastie (2021) 2.1, 3.1 and 3.4.

In statistical inference, there are two dominant paradigms: frequentist inference and Bayesian inference. Apart from the dominant paradigms, fiducial inference is also available in the literature. However, we will not discuss it in this course.

1.4.1 Frequentist Inference

Frequentist inference refers to inferential procedures whose accuracy is evaluated and guaranteed as the sample size grows to infinity or if the same data generating process is repeated infinitely many times. Here are a few examples:

Definition 1.1 (Bias)**.** The *bias* of the estimator $\hat{\theta}$ is defined as

$$Bias(\hat{\theta}) := \mathbb{E}_{\theta_0}[\hat{\theta}] - \theta_0.$$

As the bias also depends on data that are not observed, this is procedure-wise guarantee. In contrast, the actual error of estimation $\hat{\theta} - \theta_0$ is data dependent.

Example 1.4 (sample mean)**.** Suppose the data $X_1, ..., X_n \overset{iid}{\sim} F$ with mean θ_0, and the estimator for θ_0 is the sample mean:

$$\hat{\theta} = \frac{1}{n} \sum_{i=1}^{n} X_i.$$

By linearity of expectation and the iid assumption,

$$\mathbb{E}[\hat{\theta}] = \theta_0.$$

♢

Example 1.5 (sample variance)**.** Suppose the data $X_1, ..., X_n \overset{iid}{\sim} F$ with mean θ_0 and variance σ_0^2. The estimator for σ_0^2 is the sample variance:

$$\hat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X}_n)^2,$$

where $\bar{X}_n = \frac{1}{n} \sum_{i=1}^{n} X_i$. It can be shown

$$\mathbb{E}[\hat{\sigma}^2] = \sigma_0^2.$$

Given $\hat{\sigma}^2$ is unbiased, the estimator $\frac{1}{n} \sum_{i=1}^{n} (X_i - \bar{X}_n)^2$ is biased.

♢

The *Mean Squared Error (MSE)* is another measure of accuracy.

Definition 1.2 (Mean Squared Error (MSE))**.** The Mean Squared Error (MSE) is defined as

$$MSE(\hat{\theta}) := \mathbb{E}_{\theta_0}[(\hat{\theta} - \theta_0)^2].$$

It can be shown that

$$MSE(\hat{\theta}) = bias(\hat{\theta})^2 + V_{\theta_0}(\hat{\theta}).$$

MSE depends on both the bias and the variability of the estimator. Often times, estimators are faced with the tradeoff between bias and variance.

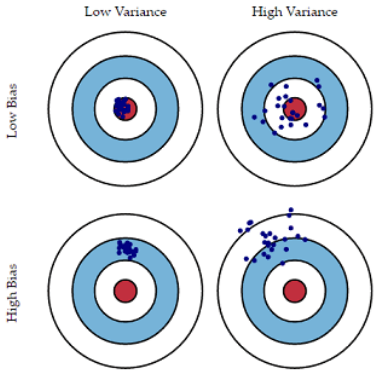


Figure 3: Bias and variance

Definition 1.3 (Consistency)**.** The (point) estimator $\hat{\theta}$ is *consistent* if

$$\hat{\theta} \to \theta_0 \text{ in probability-}\mathbb{P}_{\theta_0}.$$

The mode of convergence used above is convergence in probability. A sequence of random variables X_n converges to X in probability, written $X_n \xrightarrow{p} X$, if for every $\epsilon > 0$,

$$\mathbb{P}(|X_n - X| > \epsilon) \to 0,$$

as $n \to \infty$. A more detailed explanation of the convergence concept and relevant modes of convergence is presented in the math appendix section 1.5.

As sample size grows to infinity, the estimator gets arbitrarily close to the true value. It can be shown that $\hat{\theta}$ is consistent if its MSE goes to 0 as sample size grows to infinity.

Definition 1.4 (Rate of Convergence: Asymptotic Normality)**.** The (point) estimator $\hat{\theta}$ is *asymptotically Normal* if

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, V(\theta_0))$$

holds for some variance $V(\theta_0)$ which is potentially dependent of θ_0.

The mode of convergence used above is convergence in distribution. X_n converges to X in distribution, written $X_n \xrightarrow{d} X$, if

$$\lim_{n\to\infty} F_n(t) = F(t)$$

at all t for which F is continuous. Again, a more detailed explanation of the convergence concept and relevant modes of convergence is presented in the math appendix section 1.5.

Asymptotic Normality is often used to construct asymptotic confidence intervals/sets. We will see more details later.

Example 1.6 (Frequentist properties of sample mean)**.** Suppose the data $X_1, ..., X_n \overset{iid}{\sim} F$ with mean θ_0 and variance σ_0^2. The estimator for θ_0 is the sample mean:

$$\hat{\theta} = \frac{1}{n}\sum\nolimits_{i=1}^{n} X_i.$$

Now we can examine the frequentist properties of the estimator:

- Bias
- MSE
- Consistency
- Asymptotic Normality

Remark: the above calculations only require the first two moments of X_i. In other words, the analysis is the same for the case $X_i \overset{iid}{\sim} Ber(\theta_0)$ and the case $X_i \overset{iid}{\sim} N(\theta_0, \sigma_0^2)$ if $\sigma_0^2 = \theta_0(1-\theta_0)$. $\diamondsuit$

It is worthwhile to emphasize that the aforementioned accuracy criteria also depend on the data that are not observed by the statistician. To be more specific, bias is the *averaged* error that not merely depends on the observed data, but also potentially observed data. This argument also applies to MSE which is the *averaged* squared error. Consistency depends on more hypothetical data than the observed data.

Limit theorems like LLN and CLT describe what would happen in *asymptopia*, and there seems a gap between the theoretical guarantee and the actual situation. In fact, asymptotic effects often kick in pretty fast, and moderate amount of observed data could provide reasonably accurate inference, making frequentist criteria sensible in practice.

1.4.2 Bayesian Inference

Bayesian inference refers to *probabilistic* inferential procedures based on Bayes theorem. In contrast to frequentist inference, Bayesian inference conditions on the observed data, hence, all inferential statements are more specific to the observed data.

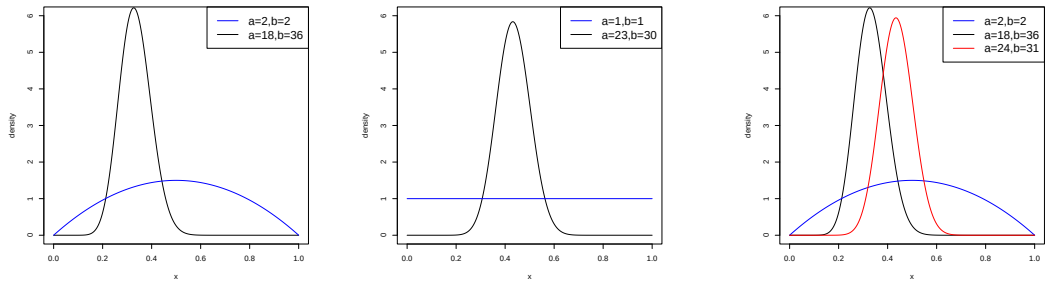
To be specific, with the probability model $\{\mathbb{P}_\theta : \theta \in \mathbb{R}\}$, we could specify a prior distribution on the parameter θ and the prior distribution admits a density function $\pi(\theta)$. With Bayes theorem, the posterior density of θ given data $\mathbf{X}$ is

$$\pi(\theta|\mathbf{X}) = \frac{\pi(\theta, \mathbf{X})}{\pi(\mathbf{X})} \propto p(\mathbf{X}|\theta)\pi(\theta)$$

where $p(\mathbf{X}|\theta)$ is the likelihood.

Example 1.7. Suppose $Y_1, ..., Y_n \overset{iid}{\sim} Ber(\theta)$ where $Y_i = 1$ denotes the success in i^{th} trial and n is the known number of trials. The model is equivalent to $X \sim Bin(n, \theta)$ where $X = \sum_{i=1}^{n} Y_i$ denotes the total number of successes. The unknown parameter is the probability of success θ.

Different prior distributions on θ produce different posterior distributions, as shown in Figure 4. Despite the difference, the posterior inferences are consistent with each other.



Prior:	$Beta(2,2)$	$Beta(1,1)$	$Beta(2,2)$
Data:	$(n, X) = (50, 16)$	$(n, X) = (51, 22)$	(50,16) and (51,22)
Posterior:	$Beta(18, 36)$	$Beta(23, 30)$	$Beta(18, 36)$ and $Beta(24, 31)$

Figure 4: From the prior to the posterior distribution

♢

Bayesian inference is made based on the posterior density $\pi(\theta|\mathbf{X})$. For instance, we can construct the point estimate using posterior mean $\mathbb{E}[\theta|\mathbf{X}] = \int \theta\pi(\theta|\mathbf{X})d\theta$, and we can construct interval/set estimate using the posterior quantiles

$$C_\alpha(\mathbf{X}) = [\pi^{-1}_{\theta|\mathbf{X}}(\alpha/2), \pi^{-1}_{\theta|\mathbf{X}}(1-\alpha/2)]$$

where $\pi^{-1}_{\theta|\mathbf{X}}(\alpha)$ denotes the αth quantile of $\pi(\theta|\mathbf{X})$. By construction, the posterior probability of $\{\theta \in C_\alpha(\mathbf{X})\}$ is $1-\alpha$. This is different from confidence intervals/sets, and it is called *credible* intervals/sets.

Some Bayesian estimators also have frequentist guarantees. For instance, Bayesian estimators can be consistent: $\mathbb{E}[\theta|\mathbf{X}] \to \theta_0$ in Probability-$\mathbb{P}_{\theta_0}$, and the credible interval can be *honest*:

$$\mathbb{P}_{\theta_0}(\theta_0 \in C_\alpha(\mathbf{X})) \geq 1-\alpha.$$

2 Descriptive Statistics and EDA

2.1 Descriptive Statistics

Definition 2.1. A statistic is a function of the data for some statistical purpose(s).

A descriptive statistic is used to summarize the sample data, e.g., the sample average. A test statistic is used in statistical hypothesis testing, e.g., reject the null hypothesis if the sample average, the test statistic, is greater than some threshold. In the examples, the sample average serves as both a descriptive statistic and a test statistic.

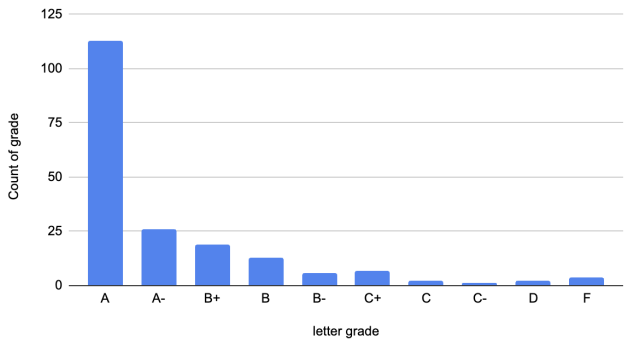
For univariate numeric data, e.g., stock prices, body weight, height, etc.,

- Maximum, minimum, range
- median, and other sample quantiles/percentiles
- sample average/mean, sample variance, sample standard deviation

For univariate discrete data, e.g., letter grades, penguin species, etc.. We can tabulate the `grades` data.

letter grade	A	A-	B	B-	B+	C	C-	C+	D	F
count	113	26	13	6	19	2	1	7	2	12

We can also plot their empirical frequency in a bar plot.




```r
library("ggplot2")
grades <- read.csv("stat131-data.csv")
head(grades)
ggplot(grades, aes(x=grade)) + geom_bar()
table(grades$grade )
```

We often prefer a nice visualization to a messy table of numbers.

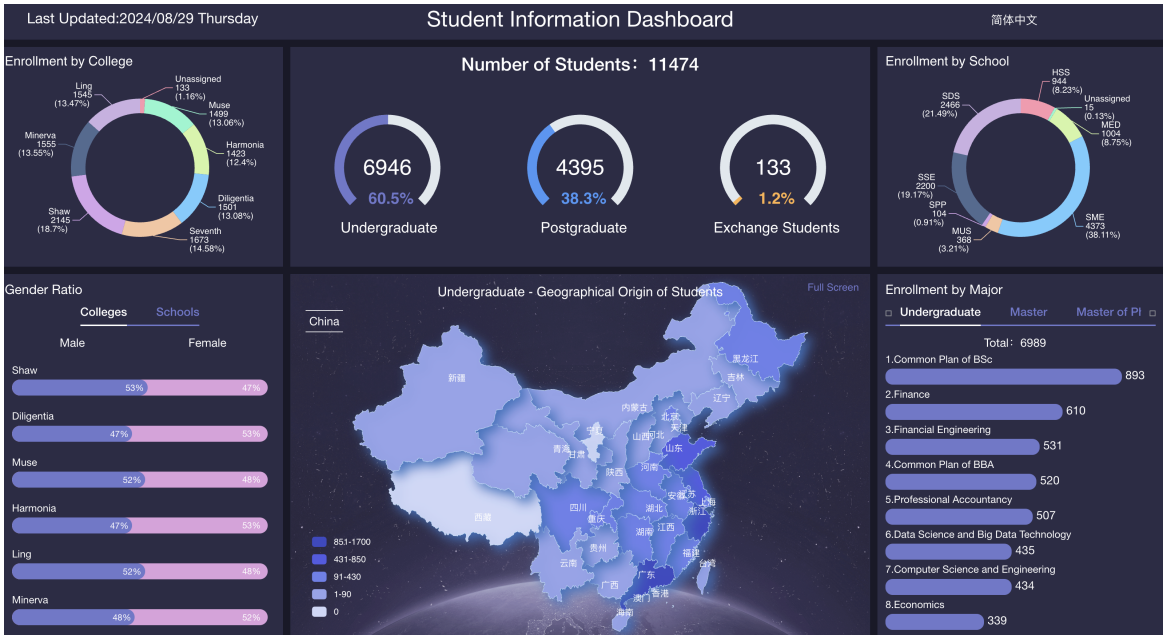


Figure 5: The student information dashboard of CUHK-SZ

Example 2.1 (Simpson's paradox)**.** Simpson's paradox refers to a phenomenon in which a trend appears in several groups of data but disappears or reverses when the groups are combined. Perhaps, the best-known example of Simpson's paradox is the study on the gender bias among graduate school admissions to University of California, Berkeley.

The following example of batting averages from the textbook Hogg et al. (2019)

	Player A			Player B		
season	AB	Hits	Average	AB	Hits	Average
1	500	126	0.252	300	75	0.250
2	300	90	0.300	500	145	0.290
totals	800	216	0.270	800	220	0.275

Ross (2007) demonstrates a similar phenomenon using the batting average of two baseball players, Derek Jeter and David Justice, during the years 1995 and 1996. In both 1995 and 1996, Justice had a higher batting average (in bold type) than Jeter did. However, when the two baseball seasons are combined, Jeter shows a higher batting average than Justice.

♢

2.2 Exploratory Data Analysis

Tukey (1977) advocated exploratory data analysis (EDA). EDA is an essential part of modern data science, as illustrated in the Box's loop.

Stem-and-leaf display Tukey (1977) introduces the stem-and-leaf display. In the first example (left of Figure 6), we simulate 10 standard Normal random numbers:

-0.626, 0.184, -0.836, 1.595, 0.330, -0.820, 0.487, 0.738, 0.576, -0.305.

After ordering and rounding, the data becomes

-0.836, -0.820, -0.626, -0.305, 0.184, 0.330, 0.487, 0.576, 0.738, 1.595.

```
The decimal point is at the |

-0 | 886
-0 | 3
 0 | 23
 0 | 567
 1 |
 1 | 6
```

10 standard Normal random numbers

```
The decimal point is 1 digit(s) to the right of the |

 0 | 2222223
 1 | 3
 2 |
 3 | 034
 4 | 6
 5 | 247
 6 | 1455668999
 7 | 0233456666677788899
 8 | 000001111222234444455555667777777788888999999999
 9 | 000000111111111222222333333444444555666666666777777777888
10 | 00000000111111122222333333444455666667888
11 | 55
```

numeric grades

Figure 6: Stem-and-leaf display

We can also "visualize" the numeric grades of the letter grades example with the stem-and-leaf display (right panel of Figure 6).

```
set.seed(1)
x <- rnorm(10)
print(x)
stem(x)
stem(grades$total)
```

The box plot A boxplot is a type of visual shorthand for measures of position (percentiles) that describe a distribution. It is also useful for identifying potential outliers.

- A box that indicates the range of the middle half of the data, a distance known as the *interquartile range* (IQR), the difference between the 25th percentile and the 75th percentile.
- Visual points that display observations that fall more than 1.5 times the IQR from either edge of the box. These are typically outliers.
- A line (or whisker) that extends from each end of the box and goes to the farthest nonoutlier point in the distribution.

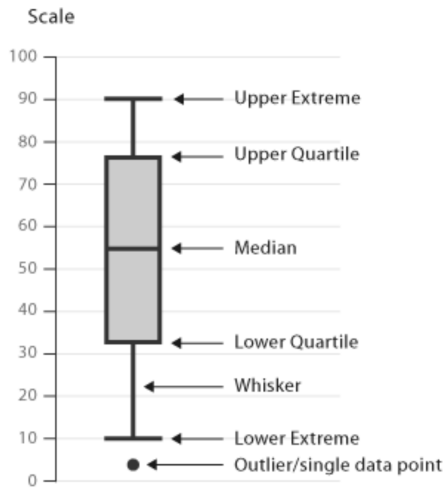


Figure 7: The composition of a box plot

Box plots are very effective in comparing the distribution of various groups. The R package `palmerpenguins` provides the great dataset `penguins` which has three species: Adelie, Chinstrap, and Gentoo.



Adelie Chinstrap Gentoo

Figure 8: The penguins

```
library(palmerpenguins)
library(tidyverse)
glimpse(penguins)
table(penguins$species)
```

Suppose we are interested in if the body masses of the species are different. We can visualize the body mass distributions by box plot.

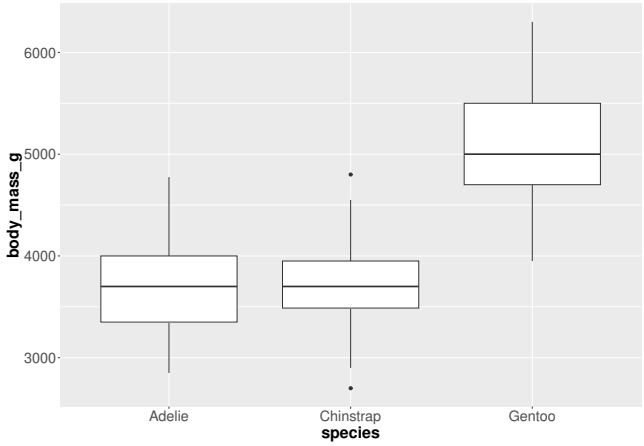


Figure 9: Compare the body mass of penguins

```
library(palmerpenguins)
library(tidyverse)
library(ggplot2)
ggplot(penguins, aes(x = species, y = body_mass_g)) + geom_boxplot()
```

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